EV CHARGING STATION MARKET ENTRY ANALYSIS

Data Analytics Project Report

Tools Used: Python, Pandas, MySQL, Power BI

1. Executive Summary

This project analyzes electric vehicle (EV) charging infrastructure data to understand geographic coverage, ownership patterns, charging infrastructure, and operator presence. Python was used for data cleaning, MySQL for data analysis, and Power BI for interactive dashboard development. The findings highlight regional differences and potential areas for further market-entry research.

2. Project Objectives

- Clean and prepare EV charging infrastructure data for analysis.
- Analyze station records, ownership, operators, and connector availability.
- Identify geographic concentration and differences in infrastructure coverage.
- Develop an interactive Power BI dashboard to communicate insights.
- Recommend areas for further market-entry investigation.

3. Tools and Methodology

Python: Extracted and cleaned the dataset using Pandas, handled missing values and duplicate records, standardized data, and validated geographic coordinates.

MySQL: Imported the cleaned dataset and used SQL queries to analyze station records, state-wise distribution, ownership, operators, districts, and connector availability.

Power BI: Developed a two-page interactive dashboard featuring KPI cards, charts, slicers, business insights, and recommendations.

4. Dataset Overview

- **Total records:** 3,039
- **Columns:** 12
- **Charging operators:** 33
- **States/UTs represented:** 8
- **Distinct districts:** 150
- **Total connectors:** 3,593

The dataset includes operator names, ownership, state, district, city/village, geographic coordinates, charger type, charger rating, connector rating, and number of connectors.

5. Key Findings

- **Geographic concentration:** Delhi recorded 883 records, followed by Andhra Pradesh with 836.
- **Ownership distribution:** Government-owned records accounted for 2,028 (66.7%), while private records accounted for 1,011 (33.3%).
- **Connector availability:** Andhra Pradesh recorded the highest connector count at 1,062, followed by Delhi with 955.
- **Operator presence:** BPCL and IOCL recorded 864 and 767 records, respectively.
- **District coverage:** West Delhi had the highest number of records, with 440.
- **Infrastructure variation:** The distribution of charger types and ownership differed across the states and districts represented in the sample.

6. Business Recommendations

1. **Investigate underserved regions** to identify potential expansion opportunities.
2. **Assess private-sector opportunities** in selected regions with limited private infrastructure.
3. **Evaluate connector requirements** against local EV demand and charging needs.
4. **Explore operator partnerships** to support infrastructure development and market entry.

7. Conclusion

The project demonstrates the use of Python, MySQL, and Power BI to transform raw data into actionable business insights. The interactive dashboard helps users explore geographic coverage, ownership distribution, operator presence, and connector availability.

The analysis provides a starting point for EV charging market-entry research. Further validation using local EV adoption, traffic patterns, competitor locations, installation costs, and site feasibility is required before making investment decisions.

7. Power BI Dashboard





Data Limitation

This analysis uses a sample of 3,039 records across 8 states/UTs. The results may not represent India's complete EV charging infrastructure, and each record is not necessarily a unique physical charging station.